Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS3DS02]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2021

SAMPLING METHODS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. i. Calculate Standard Error for the following data: 1, 9, 12, 12, 15, 18, 20. **07**

ii. Calculate the sample standard deviation of sample of 6 students to see how many books they are carrying 2, 3, 4, 2, 6, 8.

b. A population consists of four numbers 4, 7, 11, 13. Consider all possible samples of size 2 which can be drawn from this population without replacement. Find the mean and the standard deviation in the population, and in the sampling distribution of means. **07**

2. a. A biased coin is tossed 400 times and head turns up 120 times. Find the 95% confidence limits for the proportion of heads turning up in infinitely many tosses **04**

b. A box contains 10 screws out of which 3 are defective. Two screws are taken from the box. Calculate the distribution of the number of defective screws drawn. Using hyper geometric distribution. **05**

c. Suppose we select 5 cards from an ordinary deck of playing cards. Estimate the probability of obtaining 2 or fewer hearts? **05**

3. a. Refer to a process that fills boxes with a mean of 340 grams of cereal, with a standard deviation of 25 grams. Assume the probability distribution for this population is unknown. If a store purchases 600 boxes of cereal, Estimate the probability that a sample of 50 boxes from the order will average less than 328 grams? $\{\emptyset(3.42) = 0.4999\}$ **06**

b. Complaining students argue that most of the class feels this way, but Dr. Badstat is not sure. Below is a list of the 36 students in the class. **08**

1.Emily	7.Michael	13.Samantha	19.Andrew	25.Kayls	31.Megan
2.Hannah	8.Jacob	14.Jessica	20.Joy	26.Ryan	32.David
3.Madison	9.Suresh	15.Joseph	21.Daniel	27.John	33.James
4.Ashley	10.Santhosh	16.Taylor	22.William	28.Olivia	34.Emma
5.Sarah	11.Mahesh	17.Lauren	23.Tyler	29.Sandhya	35.Zachary
6.Alexis	12.Kavya	18.Alyssa	24.Brandon	30.Ani	36.Brianna

i. List the steps for selecting a simple random sample of 5 out of the 36 students

ii. Generate random Sample of 5 out of the 36 students by random number table.

OR

4. a. From the health department, it is surveyed about major rice eaters in a particular state of India, to know about the diabetes patients eating habit. It is found that out of 1200 people, 850 are major rice eaters. Estimate the confidence interval for the population proportion with 95% confidence level. **08**

b. From the population 1,2,3,4 draw samples of size two without replacement and also calculate standard error. **06**

5. a. Interpret different types of Stratified and Systematic sampling with examples. **04**

b. A Population with 12 elementary units is divided into 3 strata with values in the table given below **10**

Stratum 1	Stratum 2	Stratum 3
$X_{11} = 6$	$X_{21} = 10$	$X_{31} = 20$
$X_{12} = 4$	$X_{22} = 17$	$X_{32} = 26$
$X_{13} = 2$	$X_{23} = 12$	$X_{33} = 24$
$X_{14} = 10$		$X_{34} = 18$
$X_{15} = 8$		

Solve for σ_h^2, S_h^2 for strata samples and σ^2, S^2 for population from the given data

OR

6. a. Outline systematic sampling and delineate the steps to form a sample using the systematic sampling technique. **04**

b. Illustrate Mean, Population mean, SST, SSW, SSB, Degrees of freedom for the following group of data **10**

Group1	Group 2	Group 3	Group 4
6	8	9	10
18	19	9	6
2	5	7	4
0	5	4	7

7. a. Explain about cluster sampling with examples and delineate the conditions when to use the cluster sampling. **07**

b. The client has the following data for accounts receivable. The Audit firm will use probability proportional to size sampling. A sample size of 5 items will be selected **07**

i. Find the sampling interval

ii. Select the 5 items from the cumulative book value column

iii. Find the total projected misstatement

Item	1	2	3	4	5	6	7	8	9	10
Book Value	$500	$200	$600	$300	$350	$800	$450	$150	$750	$900
Audit value	$500	$200	$600	$300	$350	$560	$450	$150	$300	$900

OR

8. a. State difference between Cluster sampling and stratified sampling **04**

b. A population with 20000 in 30 clusters, sample 3000 from 10 clusters using Probability proportional to size. **10**

- Calculate probability 1: probability of selection for each sampled cluster
- Calculate probability 2: probability of selection for each individual in each of the sampled clusters from the following data
- Calculate the overall weight

Cluster	1	2	3	4	5	6	7	8	9	10
Size	555	1309	390	1309	698	907	432	897	677	501
Cluster	11	12	13	14	15	16	17	18	19	20
Size	867	867	1002	1094	668	500	835	396	630	483
Cluster	21	22	23	24	25	26	27	28	29	30
Size	319	569	987	598	375	387	465	751	365	448